

Yushi Zeng

Address: South China University of Technology, Guangdong, China 510006 | Tel: (+86) 132-6822-2570

Email: yushi_znn@foxmail.com | Personal Website: yushizeng.github.io

EDUCATION

South China University of Technology	Guangzhou, China
Doctor of Engineering in Future Technology (Advisor: Dr. Yi Cai)	Sept. 2023 - Present
South China University of Technology	Guangzhou, China
Master of Engineering in Software Engineering (Advisor: Dr. Yi Cai)	Sept. 2020 - Jun. 2023
Jishou University	Hunan, China
Bachelor of Engineering in Software Engineering	GPA:3.47/4.00 Sept.2016-Jun.2020

RESEARCH INTERESTS

Sentiment Analysis, Knowledge Graph, Federate Learning and Multimodal learning.

PUBLICATIONS

- [1] **Zeng, Y.**, Ren, H., Cai, Y., Li, Y., Qin, J., & Wang, Y. (2026). Modality augmentation and task-aware dual-modal LoRAs for multi-task multimodal federated learning. *Information Processing & Management*, 63(4), 104601. (**Accepted; SCI; IF/JCR:8.1/Q1; First Author**)
- [2] Li, Y., Zeng, Y., Jiang, D., Wang, Y., & Lu, G. (2026). Rethinking the Knowledge Gap between Cloud and Device Models for Effective Co-Adaptation. *IEEE Transactions on Circuits and Systems for Video Technology*.(**Accepted; SCI; IF/JCR:10.8/Q1; Co-Author**)
- [3] Zeng, Y., Ren, H., Cai, Y., Li, Y., Wang, Y., & Li, Q. (2026). Class-Incremental Cloud-Device Collaborative Adaptation With Contrastive Learning in Dynamic Changing Environments. *IEEE Transactions on Neural Networks and Learning Systems*. (**Accepted; SCI; IF/JCR:9.1/Q1; First Author**)
- [4] Ren, H., Zeng, Y., Cai, Y., Ye, Z., Yuan, L., & Zhu, P. (2024, May). Grounded multimodal procedural entity recognition for procedural documents: a new dataset and baseline. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)* (pp. 7971-7981).(**Accepted; CCF-B International Conference; Co-Author**)
- [5] Zeng, Y., Wang, G., Ren, H., Cai, Y., Leung, H. F., Li, Q., & Huang, Q. (2023). A knowledge-enhanced and topic-guided domain adaptation model for aspect-based sentiment analysis. *IEEE Transactions on Affective Computing*, 15(2), 709-721. (**Accepted; SCI; IF/JCR:11.3/Q1; First Author**)
- [6] Ren, H., Zeng, Y., Cai, Y., Zhou, B., & Lian, Z. (2023, July). Constructing procedural graphs with multiple dependency relations: A new dataset and baseline. In *Findings of the Association for Computational Linguistics: ACL 2023* (pp. 8474-8486).(**Accepted; CCF-A International Conference; Co-Author**)
- [7] Zeng, Y., Wang, G., Ren, H., & Cai, Y. (2022, June). Enhance cross-domain aspect-based sentiment analysis by incorporating commonsense relational structure (Student Abstract). In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 36, No. 11, pp. 13105-13106).(**Accepted; CCF-A International Conference; First Author**)

RESEARCH EXPERIENCE

Research on Standard knowledge graph construction and application

Research Member

Jingshan WPS

Sept 2020 — Nov 2021

- Knowledge Graph Construction: The Schema construction for the standard domain and the knowledge extraction Extract the entity-relation triples by utilizing the inherent document structure features of the standard documents. Moreover, the BILSTM-CRF and BERT-CRF models are used to extract the fine-grained entities. Then, the extracted standard knowledge triples are stored in the neo4j database and we conduct the visualization analysis.
- tandard Knowledge Recommendation: We adopt a content-based recommendation method which uses the user-entered keywords to retrieve and link the relevant entities from standard knowledge graph.
- Standard conflict detection: We recognize the candidate entity from the standard drafts and kink them from the knowledge graph and identify the suspected conflicting entries through the text comparison.

INTERNSHIP EXPERIENCE

Huawei | “AI Applications in Wireless Information Experience”

IT Platform Department 3

Shenzhen, China

Jun. 2022 — Sep. 2022

- Worked on an internal Huawei project involving heterogeneous knowledge bases, solving cross-graph retrieval challenges by linking product and terminology knowledge graphs to improve the relevance of technical entity retrieval.
- Built an entity recognition, linking, and disambiguation pipeline with Jieba, pre-trained BERT embeddings, and cosine similarity, reducing ambiguity in product names and attribute descriptions and improving matching accuracy.
- Improved system deployment and API performance by designing a multi-threaded access strategy, raising entity linking accuracy to over 90% and ensuring 99% of response times remained under 200 ms.

AWARDS & COMPETITION

- Wuliangye Outstanding Student Scholarship 2023
- China National Scholarship 2022
- First-Class Postgraduate Scholarship 2022
- AAAI Student Scholarship Program 2022
- Certificate of Completion, Summer Research Experience Programme for Mainland and Taiwan Students, The Chinese University of Hong Kong 2022
- Outstanding Graduate, Jishou University 2020
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- First-Class and Second-Class Scholarships, Jishou University (Awarded over Four Consecutive Years) 2017–2020
- Third Prize, The 10th Lanqiao Cup National Software and Information Technology Competition 2019

SKILLS & INTERESTS

- **Languages:** Mandarin Chinese, English (Fluent, CET6)
- **Programming Languages:** Python (PyTorch), JAVA, C
- **Software and Tools:** Microsoft Office (Word, Excel, and PowerPoint), Zotero, LaTeX, Markdown
- **Interests:** Badminton, Fitness